《东大高数速成课》
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2015-2016 学年期末考试试卷参考答案

一、 解答下列各题(每题 6 分，共 54 分)

1、【解析】设$F(x)=f(x)+g(x)$，则

$$F(x)=\begin{cases}x+b, & x<0\\ 2x+2, & 0\leqslant x<1\\ x+2+a, & x\geqslant 1\end{cases}，由F(0^+)=F(0^-),F(1^+)=F(1^-)得\begin{cases}b=2\\ 4=3+a\end{cases},所以a=1,b=2$$

2、【解析】$\lim\limits_{x\to 0}\dfrac{(x-2)e^x+x+2}{\sin^3 x}=\lim\limits_{x\to 0}\dfrac{(x-2)e^x+x+2}{x^3}$

$$=\lim_{x\to 0}\frac{(x-1)e^x+1}{3x^2}=\lim_{x\to 0}\frac{xe^x}{6x}=\frac{1}{6}$$

3、【解析】

方法一：原积分$=\int\dfrac{\left(\sqrt{x+1}-1\right)^2}{\left(\sqrt{x+1}+1\right)\left(\sqrt{x+1}-1\right)}dx=\int\left(1+\dfrac{2}{x}-\dfrac{2\sqrt{x+1}}{x}\right)dx$

$=x+2\ln|x|-\int\dfrac{2\sqrt{x+1}}{x}dx\ \ \underline{\underline{t=\sqrt{x+1}}}\ x+2\ln|x|-4\int\dfrac{t^2}{t^2-1}dt$

$=x+2\ln|x|-4\int\left[1+\dfrac{1}{2}\left(\dfrac{1}{t-1}-\dfrac{1}{t+1}\right)\right]dt=x+2\ln|x|-4t+\ln\left|\dfrac{t+1}{t-1}\right|+C$

$=x+2\ln|x|-4\sqrt{x+1}+4\ln\left|1+\sqrt{x+1}\right|-2\ln|x|+C=x-4\sqrt{x+1}+4\ln\left|1+\sqrt{x+1}\right|+C$

方法二：（复杂）设$\sqrt{x+1}=\tan^2 t\left(0\leqslant t<\dfrac{\pi}{2}\right)\therefore x=\tan^4 t-1,dx=4\tan^3 t\sec^2 t dt$

$\therefore$原式$=\int\dfrac{\tan^2 t-1}{\tan^2 t+1}4\tan^3 t\sec^2 t dt=4\int\tan^5 t dt-4\int\tan^3 t dt$

$\because\int\tan^n t dt=\int\tan^{n-2}t(\sec^2 t-1)dt=\int\tan^{n-2}t d\tan t-\int\tan^{n-2}t dt$

$\therefore\int\tan^3 t dt=\dfrac{\tan^2 t}{2}+\ln\cos t+C_1\quad\int\tan^5 t dt=\dfrac{\tan^4 t}{4}-\dfrac{\tan^2 t}{2}-\ln\cos t+C_2$

$\therefore$原式$=\tan^4 t-2\tan^2 t+2\ln|1+\tan^2 t|-2\tan^2 t+2\ln|1+\tan^2 t|+C_3$

$=x-4\sqrt{x+1}+4\ln\left|1+\sqrt{1+x}\right|+C$

4、【解析】令$t=\dfrac{\sin x-1}{\sin x+1}$

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$\therefore dt=\dfrac{\cos x(\sin x+1)-(\sin x-1)\cos x}{(\sin x+1)^2}dx=\dfrac{2\cos x}{(\sin x+1)^2}dx$

$dy=df(t)=\arctan t^2dt=\arctan\left(\dfrac{\sin x-1}{\sin x+1}\right)^2\dfrac{2\cos x}{(\sin x+1)^2}dx\therefore\left.\dfrac{dy}{dx}\right|_{x=0}=\dfrac{\pi}{2}$

5、【解析】设$t=\arccos x$，则$x=\cos t,\ dx=-\sin tdt$

$\therefore$原式$=\displaystyle\int_{\frac{\pi}{2}}^{0}t^2(-\sin t)dt=\cos t\cdot t^2\Big|_{\frac{\pi}{2}}^{0}-\int_{\frac{\pi}{2}}^{0}\cos t\cdot 2tdt$

$=-\sin t\cdot 2t\Big|_{\frac{\pi}{2}}^{0}+\displaystyle\int_{\frac{\pi}{2}}^{0}2\sin tdt=\pi-2$

6、【解析】

（1）$b_3=\dfrac{1}{\pi}\displaystyle\int_{-\pi}^{\pi}f(x)\sin 3xdx=\dfrac{1}{\pi}\int_{-\pi}^{\pi}x\sin 3xdx=\dfrac{1}{3}$

（2）该函数满足收敛定理的条件。它在$x=(2k+1)\pi(k\in Z)$处不连续，其他地方连续

$\therefore\sigma(81\pi)=\dfrac{f(\pi^-)+f(-\pi^+)}{2}=\dfrac{\pi+0}{2}=\dfrac{\pi}{2}\quad\sigma\left(\dfrac{26\pi}{3}\right)=\sigma\left(\dfrac{26\pi}{3}-8\pi\right)=\sigma\left(\dfrac{2\pi}{3}\right)=\dfrac{2\pi}{3}$

7、【解析】$\because f(0^+)=1,f(0^-)=1\ \therefore f(x)$在$x=0$处连续，设$t=x-1$

$\therefore\displaystyle\int_0^2f(x-1)dx=\int_0^2f(x-1)d(x-1)=\int_{-1}^{1}f(t)dt=\int_{-1}^{0}\frac{1}{1+e^t}dt+\int_0^1\frac{1}{t+1}dt$

$\displaystyle\int_{-1}^{0}\left(1-\frac{e^t}{1+e^t}\right)dt+\ln(1+t)\Big|_0^1=[t-\ln(1+e^t)]\Big|_{-1}^{0}+\ln 2$

$=-\ln 2+\ln(1+e)+\ln 2=\ln(1+e)$

8、【解析】$f'(x)=xe^{-x^2}\ \therefore x>0$时$f'(x)>0,x<0$时$f'(x)<0$

$\therefore f(x)$在$(-\infty,0)$递减，$(0,+\infty)$递增

$\therefore f(x)$极小值为$f(0)=0$，无极大值

$\because f''(x)=e^{-x^2}-2x^2e^{-x^2}=(1-2x^2)e^{-x^2}=0\Leftrightarrow x=\pm\dfrac{\sqrt{2}}{2}$

$x\in\left(-\infty,-\dfrac{\sqrt{2}}{2}\right)$时，$f''(x)<0;x\in\left(-\dfrac{\sqrt{2}}{2},\dfrac{\sqrt{2}}{2}\right)$时，$f''(x)>0;x\in\left(\dfrac{\sqrt{2}}{2},+\infty\right)$时，$f''(x)<0,$

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∴拐点横坐标是$x=\pm\frac{\sqrt{2}}{2}$.

9、【解析】方程两边同时对x求导得$\frac{1}{2}\frac{2x+2y\frac{dy}{dx}}{x^2+y^2}=\frac{\frac{\frac{dy}{dx}x-y}{x^2}}{1+\left(\frac{y}{x}\right)^2}\Rightarrow\frac{dy}{dx}=\frac{x+y}{x-y}$

$\therefore\frac{d^2y}{dx^2}=\frac{\left(1+\frac{dy}{dx}\right)(x-y)-(x+y)\left(1-\frac{dy}{dx}\right)}{(x-y)^2}=\frac{2x+\frac{2y(x+y)}{x-y}}{(x-y)^2}=\frac{2x^2+2y^2}{(x-y)^3}$

二、 **(9 分)**

【解析】不妨设高$h\geqslant1$，底面半径为r，则$r^2+(h-1)^2=1\Rightarrow r^2=2h-h^2$

$\therefore V=\frac{1}{3}\pi r^2h=\frac{\pi}{3}(2h-h^2)h=\frac{\pi}{3}(2h^2-h^3)$，

$V'=\frac{\pi}{3}(4h-3h^2)$，令$V'=0$，得$h=\frac{4}{3}$，在$\left(1,\frac{4}{3}\right)$上，$V'>0,V$递增；在$\left(\frac{4}{3},+\infty\right)$上，

$V'<0,V$递减.所以当$h=\frac{4}{3}$时，V取得最大值$V_{\max}=\frac{32\pi}{81}$.

三、 **计算题（9 分）**

【解析】易知OA方程为$y=ex,A(1,e)$

（1）面积$S=\int_{-\infty}^{1}e^xdx-\int_0^1exdx=e^x\Big|_{-\infty}^{1}-\frac{ex^2}{2}\Big|_0^1=\frac{e}{2}$

（2）体积$V=\int_{-\infty}^{0}\pi e^{2x}dx+\int_0^1(\pi e^{2x}-\pi e^2x^2)dx=\frac{\pi e^{2x}}{2}\Big|_{-\infty}^{0}+\left(\frac{\pi e^{2x}}{2}-\frac{\pi e^2x^3}{3}\right)\Big|_0^1=\frac{\pi e^2}{6}$

四、 **（9 分）**【解析】$\because\lim\limits_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim\limits_{n\to\infty}\frac{n^2+n+1}{n^2-n+1}=1$且$x=\pm1$时级数发散，所以收敛域为$(-1,1)$，

$\therefore\sum_{n=2}^{\infty}(n^2-n+1)x^n=\sum_{n=1}^{\infty}n^2x^n+\sum_{n=1}^{\infty}nx^n+\sum_{n=0}^{\infty}x^n-x-1$

$=\frac{x(1+x)}{(1-x)^2}-\frac{x}{(1-x)^2}+\frac{1}{1-x}-x-1=\frac{2x^2-x^3}{(1-x)^2}$，令$x=-\frac{1}{2}$即得和为$\frac{5}{18}$.

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五、 （9 分）【解析】$\because f'(x)=\dfrac{-1}{1-x}=-\sum\limits_{n=0}^{\infty}x^n,\therefore f(x)=-\sum\limits_{n=0}^{\infty}\dfrac{x^{n+1}}{n+1}=-\sum\limits_{n=1}^{\infty}\dfrac{x^n}{n}$

$\therefore I=-\int_0^1\sum\limits_{n=1}^{\infty}\dfrac{x^{n-1}}{n}dx=-\sum\limits_{n=1}^{\infty}\dfrac{x^n}{n^2}\Big|_0^1=-\sum\limits_{n=1}^{\infty}\dfrac{1}{n^2}=-\dfrac{\pi^2}{6}$

六、 证明题（每小题 5 分，共 10 分）

1、【证明】由积分中值定理得$\exists a\in(-1,0)$使$af(a)=\int_{-1}^{0}xf(x)dx>0\Rightarrow f(a)<0$

$\exists b\in(0,1)$使$\sin bf(b)=\int_0^1\sin xf(x)dx>0\Rightarrow f(b)>0$

$\because f(a)f(b)<0\ \therefore\exists\eta\in(a,b)$使$f(\eta)=0$又$f(2)=0$，

由罗尔定理得，$\exists\xi\in(\eta,2)\subseteq(-1,2)$使$f'(\xi)=0$

2、【证明】$\int_0^{+\infty}\dfrac{\sin^2x}{x^2}dx=-\dfrac{\sin^2x}{x}\Big|_0^{+\infty}-\int_0^{+\infty}-\dfrac{2\sin x\cos x}{x}dx=\int_0^{+\infty}\dfrac{\sin 2x}{2x}d2x=\dfrac{\pi}{2}$

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